RV College of Engineering®
Mysore Road, RV Vidyaniketan Post,
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Department of Artificial Intelligence and Machine Learning

Date: 10.011.2025	CIE 1	Max. Marks: 50+10
Semester: V	UG	Duration: $1\frac{1}{2}$ Hrs + ½ Hr
Title: Mathematical Algorithms for Artificial Intelligence		Course Code: AI255BTC

Part A

Q. No		PART A Quiz 1	M	BT	CO
1	a)	**Statement 1**: A system of linear equations is consistent if it has a unique solution. **Statement 2:** A system of linear equations is inconsistent if it has infinitely many solutions. Identify the correct option with respect to a consistent system. (a) **ONLY** Statement 1 is **TRUE** (b) **ONLY** Statement 2 is **TRUE** (c) Both Statement 1 and Statement 2 are TRUE (d) Both Statement 1 and Statement 2 are FALSE	2	1	1
	b)	Which among the following is **NOT** an algebraic field? (a) R_{57} the set of reminders modulo 57 (b) R_{53}, the set of reminders modulo 53 (c) R_{47}, the set of reminders modulo 47 (d) R_{61}, the set of reminders modulo 61 (e) All of them are fields.	2	1	2
	c)	Which among the following is not a vector space over $\mathbb{R}$, the field of real numbers, with real number addition and multiplication operations (a) $\mathbb{R}^2$ (b) $\mathcal{S}_1 = \left\{ \begin{pmatrix} 2 \\ 3 \end{pmatrix} \right\}$ (c) $\mathcal{S}_2 = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} : x_2 = 0 \right\}$ (d) $\mathcal{S}_3 = \left\{ \begin{pmatrix} x_1 \\ x_2 \end{pmatrix} : x_2 = 4x_1 \right\}$	2	2	1
	d)	You are given the following statements: **Statement 1: A vector space is a linearly dependent set** **Statement 2: A basis for a vector space is a linearly independent set.** (a) Both statements are FALSE (b) Both statements are TRUE (c) Statement 1 is ALWAYS FALSE, STATEMENT 2 IS ALWAYS TRUE (d) Statement 1 is ALWAYS TRUE, STATEMENT 2 is ALWAYS FALSE	2	2	2
	e)	Which of the following is NOT TRUE about a set containing only one non-zero vector? (a) The set is a linearly independent set (b) The set is a basis for a 1-D subspace (c) The set is a vector space over a field of real numbers (d) The span of this set is a line passing through the origin	2	2	2

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Q. No	PART B CIE	M	B	CO
1	Let V be a real vector space. Suppose that $\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3, \mathbf{v}_4$ are vectors in V which are linearly independent. Prove or Disprove the vectors $\mathbf{v}_1, \mathbf{v}_1+\mathbf{v}_2, \mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3, \mathbf{v}_1+\mathbf{v}_2+\mathbf{v}_3+\mathbf{v}_3$ are also linearly independent.	6	2	1
2	Let $\begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ have non-zero solution $\mathbf{x}_h$ to the corresponding homogeneous system of equations and $\mathbf{x}_p$ be a particular solution for the given **b (b ≠ the zero vector). Prove or disprove that the complete set of solutions** $\{\mathbf{x}_{complete} = \mathbf{x}_p + \alpha \mathbf{x}_h\}$ is a vector subspace of $\mathbb{R}^2$, for real scalars α.	6	2	1
3	Assert or Reject each of the following 3 statements with proper justification. **Statement 1:** The set containing a single vector is **ALWAYS** a linearly **DEPENDENT** set. **Statement 2:** The set containing a single vector is **ALWAYS** a basis for some vector subspace. **Statement 3:** Every vector space contains at least one subspace.	6	3	2
4	Determine if the following statements are **TRUE** or **FALSE. Justify your answer with an example or a counterexample.** **(a)** Every non-zero subspace of say $\mathbb{R}^5$ has infinitely many bases. (b) The set of columns of a 3x3 matrix will form a basis for $\mathbb{R}^3$. (c) The rank of a matrix A is the number of non-zero rows in it. (d) The vector space $\mathbb{R}^{27}$ contains 27 subspaces.	12	3	2
5	Find the condition under which the following system of equations is consistent. $3x_1 + x_2 + x_3 = b_1$ $x_1 - x_2 + 2x_3 = b_2$ $2x_1 - 2x_2 + 4x_3 = b_3$	10	3	2
6	Let the set $\mathbb{R}^3$ with operations $\oplus$ and $\odot$ defined as follows. For any two vectors $\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix}, \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ y_3 \end{pmatrix}$, in the set, vector addition is defined as $\mathbf{x} \oplus \mathbf{y} = \begin{pmatrix} x_1 + y_1 + 5 \\ x_2 + y_2 - 7 \\ x_3 + y_3 + 1 \end{pmatrix}$ For a real scalar k, scalar multiplication is defined as $k \odot \mathbf{x} = \begin{pmatrix} kx_1 + 5(k-1) \\ kx_2 - 7(k-1) \\ kx_3 + k - 1 \end{pmatrix}$ Find the zero vector for this vector space with the operations as defined above.	10	3	2